

MATH 3260 - Chapter 5 Notes: Inner Product Spaces

Feedback-Revised Student Notes

Kennesaw State University - MATH 3260

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These notes are synchronized with the feedback-revised student notebook. They emphasize definitions, theorems, worked mathematics, conceptual connections, and interpretation while omitting notebook-specific code and worksheet instructions.

Chapter overview

1. **Section 5.1:** Length and Dot Product in $\mathbb{R}^n$
2. **Section 5.2:** Inner Product Spaces
3. **Section 5.3:** Orthonormal Bases and the Gram-Schmidt Process
4. **Section 5.4:** Mathematical Models and Least Squares Analysis
5. **Section 5.5:** Applications of Inner Product Spaces

Each section ends with a terminology table, a worksheet, and complete instructor solutions.

Original-example policy

All worked examples and numerical values are original. Themes are used selectively from the San Antonio Spurs, hiking, badminton, driving, travel, Backstreet Boys, and R&B music. Any prices, routes, coordinates, listening data, or performance data are fictional teaching models unless explicitly identified otherwise.

Section 5.1 - Length and Dot Product in $\mathbb{R}^n$

Learning objectives

Students should be able to:

- compute vector length and construct unit vectors;
- compute distances between vectors;
- use the dot product and its algebraic properties;
- find angles and test orthogonality;
- apply Cauchy-Schwarz, the triangle inequality, and the Pythagorean theorem;
- express the dot product as a matrix product.

Definition: norm

For $\mathbf{v} = (v_1, \dots, v_n)$, the Euclidean norm is $\|\mathbf{v}\| = \sqrt{v_1^2 + \dots + v_n^2}$. A vector with norm 1 is a unit vector.

Spurs-inspired norm example

Let

$$\mathbf{v} = (9, 12, 20).$$

Then

$$\|\mathbf{v}\| = \sqrt{9^2 + 12^2 + 20^2} = \sqrt{81 + 144 + 400} = \sqrt{625} = 25.$$

The unit vector in the direction of $\mathbf{v}$ is

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|} = \left(\frac{9}{25}, \frac{12}{25}, \frac{4}{5} \right).$$

Distance

The Euclidean distance between $\mathbf{u}$ and $\mathbf{v}$ is $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$.

Travel-coordinate distance

Suppose two fictional travel profiles are encoded by

$$\mathbf{u} = (3, -2, 5), \quad \mathbf{v} = (9, 1, -3).$$

Then

$$\mathbf{u} - \mathbf{v} = (-6, -3, 8),$$

so

$$d(\mathbf{u}, \mathbf{v}) = \sqrt{(-6)^2 + (-3)^2 + 8^2} = \sqrt{109}.$$

Definition: dot product

For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, $\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + \cdots + u_n v_n$.

Dot product and angle

Let

$$\mathbf{a} = (4, 1, -2), \quad \mathbf{b} = (2, -3, 5).$$

The dot product is

$$\mathbf{a} \cdot \mathbf{b} = 4(2) + 1(-3) + (-2)(5) = -5.$$

Also,

$$\|\mathbf{a}\| = \sqrt{21}, \quad \|\mathbf{b}\| = \sqrt{38}.$$

Therefore,

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} = -\frac{5}{\sqrt{798}}.$$

For nonzero vectors,

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}.$$

Thus

$$\mathbf{a} \perp \mathbf{b} \iff \mathbf{a} \cdot \mathbf{b} = 0,$$

which corresponds to a 90° angle.

Orthogonality

Two vectors are orthogonal exactly when their dot product is zero.

Badminton movement example

Consider two idealized movement vectors on a badminton court:

$$\mathbf{p} = (3, 4, -1), \quad \mathbf{q} = (4, -3, 0).$$

Because

$$\mathbf{p} \cdot \mathbf{q} = 3(4) + 4(-3) + (-1)(0) = 0,$$

the vectors are orthogonal.

Cauchy-Schwarz, triangle inequality, and Pythagorean theorem

For vectors $\mathbf{u}$ and $\mathbf{v}$:

$$|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|,$$

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|.$$

If $\mathbf{u} \perp \mathbf{v}$, then

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2.$$

Matrix form of the dot product

Treat vectors as columns. Then

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v}.$$

For example,

$$\begin{bmatrix} 4 & 1 & -2 \end{bmatrix} \begin{bmatrix} 2 \\ -3 \\ 5 \end{bmatrix} = -5.$$

Closing note

The dot product connects algebra and geometry: it measures directional alignment, determines angles, and detects orthogonality.

Definitions and terminology

Term	Meaning
Norm or length	$\ \mathbf{v}\ = \sqrt{\mathbf{v} \cdot \mathbf{v}}$.
Unit vector	Vector of length 1.
Distance	$\ \mathbf{u} - \mathbf{v}\ $.
Dot product	Sum of componentwise products.
Orthogonal	Dot product equals zero.
Angle	Determined by the normalized dot product.
Cauchy-Schwarz inequality	$ \mathbf{u} \cdot \mathbf{v} \leq \ \mathbf{u}\ \ \mathbf{v}\ $.
Triangle inequality	$\ \mathbf{u} + \mathbf{v}\ \leq \ \mathbf{u}\ + \ \mathbf{v}\ $.

Section 5.2 - Inner Product Spaces

Learning objectives

Students should be able to:

- verify the inner-product axioms;
- compute inner products in vector, matrix, polynomial, and function spaces;
- use induced length, distance, angle, and orthogonality;
- find orthogonal projections in general inner product spaces.

Definition: inner product

An inner product $\langle \mathbf{u}, \mathbf{v} \rangle$ is symmetric, linear in one argument, and positive definite. A vector space equipped with an inner product is an inner product space.

The four real inner-product axioms

For all $\mathbf{u}, \mathbf{v}, \mathbf{w}$ and scalar c :

$$\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle,$$

$$\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle,$$

$$\langle c\mathbf{u}, \mathbf{v} \rangle = c\langle \mathbf{u}, \mathbf{v} \rangle,$$

and

$$\langle \mathbf{v}, \mathbf{v} \rangle \geq 0,$$

with equality only for $\mathbf{v} = \mathbf{0}$.

A proposed inner product must satisfy **all four** axioms. One failed axiom is enough to reject the operation.

Weighted R&B feature inner product

Suppose a fictional song feature vector is (r, b) , where the two components represent two normalized audio features. Define

$$\langle \mathbf{u}, \mathbf{v} \rangle_W = 2u_1v_1 + 5u_2v_2.$$

The positive weights make both coordinate contributions nonnegative in

$$\langle \mathbf{v}, \mathbf{v} \rangle_W = 2v_1^2 + 5v_2^2.$$

For

$$\mathbf{u} = (3, -1), \quad \mathbf{v} = (2, 4),$$

we obtain

$$\langle \mathbf{u}, \mathbf{v} \rangle_W = 2(3)(2) + 5(-1)(4) = -8.$$

The induced norm of $\mathbf{u}$ is

$$\|\mathbf{u}\|_W = \sqrt{2(3)^2 + 5(-1)^2} = \sqrt{23}.$$

For a weighted dot product, the weights should be positive. Zero or negative weights can destroy positive definiteness.

A form that fails

The expression $u_1v_1 - u_2v_2$ is not an inner product on $\mathbb{R}^2$, because $\langle (0, 1), (0, 1) \rangle = -1 < 0$.

Inner product on polynomial coefficient vectors

For

$$p(x) = a_0 + a_1x + a_2x^2, \quad q(x) = b_0 + b_1x + b_2x^2,$$

define

$$\langle p, q \rangle = a_0b_0 + 2a_1b_1 + 3a_2b_2.$$

Let

$$p(x) = 1 + 2x - x^2, \quad q(x) = 3 - x + 2x^2.$$

Then

$$\langle p, q \rangle = 1(3) + 2(2)(-1) + 3(-1)(2) = -7.$$

Integral inner product on functions

On $C[a, b]$, a common inner product is

$$\langle f, g \rangle = \int_a^b f(x)g(x) dx.$$

For $f(x) = 1$ and $g(x) = x$ on $[0, 1]$,

$$\langle f, g \rangle = \int_0^1 x dx = \frac{1}{2}.$$

Projection formula

For $\mathbf{v} \neq \mathbf{0}$, $\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\langle \mathbf{v}, \mathbf{v} \rangle} \mathbf{v}$.

Orthogonal projection in $\mathbb{R}^3$

Let

$$\mathbf{u} = (6, 3, 3), \quad \mathbf{v} = (1, 2, 2).$$

Since

$$\mathbf{u} \cdot \mathbf{v} = 18, \quad \mathbf{v} \cdot \mathbf{v} = 9,$$

we have

$$\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{18}{9} \mathbf{v} = 2(1, 2, 2) = (2, 4, 4).$$

The orthogonal residual is

$$\mathbf{u} - \text{proj}_{\mathbf{v}} \mathbf{u} = (4, -1, -1),$$

and

$$(4, -1, -1) \cdot (1, 2, 2) = 0.$$

Projection minimizes distance

Every vector of the form $c\mathbf{v}$ lies on the line spanned by $\mathbf{v}$. The projection is the unique point on that line closest to $\mathbf{u}$.

Closing note

An inner product equips an abstract vector space with geometry: length, angle, orthogonality, and projection.

Definitions and terminology

Term	Meaning
Inner product	Scalar-valued function satisfying symmetry, linearity, and positive definiteness.
Inner product space	Vector space with a specified inner product.
Induced norm	$\ \mathbf{v}\ = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$.
Orthogonal	Inner product equals zero.
Weighted inner product	Coordinates contribute with positive weights.
Function inner product	Often an integral of a product.
Orthogonal projection	Closest vector in a one-dimensional subspace.
Residual	Difference between a vector and its projection.

Section 5.3 - Orthonormal Bases and the Gram-Schmidt Process

Learning objectives

Students should be able to:

- distinguish orthogonal and orthonormal sets;
- prove that an orthogonal nonzero set is linearly independent;
- find coordinates relative to an orthonormal basis;
- apply the Gram-Schmidt orthonormalization process;
- construct an orthonormal basis for a solution space.

Orthogonal and orthonormal sets

A set is orthogonal when distinct vectors have inner product zero. It is orthonormal when it is orthogonal and every vector has norm one.

A nonstandard orthonormal basis for $\mathbb{R}^3$

Consider

$$\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1, 1, 0), \quad \mathbf{u}_2 = \frac{1}{\sqrt{2}}(1, -1, 0), \quad \mathbf{u}_3 = (0, 0, 1).$$

The pairwise dot products are zero, and each vector has norm one. Therefore the set is an orthonormal basis of $\mathbb{R}^3$.

Verify explicitly that all pairwise dot products are zero and that

$$\|\mathbf{u}_1\| = \|\mathbf{u}_2\| = \|\mathbf{u}_3\| = 1.$$

Coordinates relative to an orthonormal basis

For

$$\mathbf{w} = (9, 12, 20),$$

the coordinates are simply the inner products:

$$\begin{aligned} c_1 &= \mathbf{w} \cdot \mathbf{u}_1 = \frac{21}{\sqrt{2}}, \\ c_2 &= \mathbf{w} \cdot \mathbf{u}_2 = -\frac{3}{\sqrt{2}}, \\ c_3 &= \mathbf{w} \cdot \mathbf{u}_3 = 20. \end{aligned}$$

Thus

$$[\mathbf{w}]_B = \begin{bmatrix} 21/\sqrt{2} \\ -3/\sqrt{2} \\ 20 \end{bmatrix}.$$

Gram-Schmidt idea

Subtract previously constructed projections to create orthogonal vectors, then normalize each vector.

Goal: start with a linearly independent set and produce an orthonormal basis for the same span. Remove projections onto directions already constructed, then normalize.

Detailed Gram-Schmidt example in $\mathbb{R}^3$

Start with

$$\mathbf{v}_1 = (1, 1, 0), \quad \mathbf{v}_2 = (1, 0, 1), \quad \mathbf{v}_3 = (0, 1, 1).$$

First,

$$\mathbf{w}_1 = \mathbf{v}_1 = (1, 1, 0).$$

For the second vector,

$$\text{proj}_{\mathbf{w}_1} \mathbf{v}_2 = \frac{\mathbf{v}_2 \cdot \mathbf{w}_1}{\mathbf{w}_1 \cdot \mathbf{w}_1} \mathbf{w}_1 = \frac{1}{2}(1, 1, 0),$$

so

$$\mathbf{w}_2 = \mathbf{v}_2 - \text{proj}_{\mathbf{w}_1} \mathbf{v}_2 = \left(\frac{1}{2}, -\frac{1}{2}, 1\right).$$

Scaling does not change orthogonality, so use the simpler vector

$$\mathbf{w}_2 = (1, -1, 2).$$

For the third vector,

$$\text{proj}_{\mathbf{w}_1} \mathbf{v}_3 = \frac{1}{2}(1, 1, 0),$$

and

$$\text{proj}_{\mathbf{w}_2} \mathbf{v}_3 = \frac{1}{6}(1, -1, 2).$$

Therefore

$$\mathbf{v}_3 - \text{proj}_{\mathbf{w}_1} \mathbf{v}_3 - \text{proj}_{\mathbf{w}_2} \mathbf{v}_3 = \frac{2}{3}(-1, 1, 1).$$

Use

$$\mathbf{w}_3 = (-1, 1, 1).$$

After normalization,

$$\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1, 1, 0), \quad \mathbf{u}_2 = \frac{1}{\sqrt{6}}(1, -1, 2), \quad \mathbf{u}_3 = \frac{1}{\sqrt{3}}(-1, 1, 1).$$

Gram-Schmidt on a solution space

Consider

$$-x_1 + x_2 + x_3 = 0, \quad x_1 + x_4 = 0.$$

Solving gives a basis

$$\mathbf{v}_1 = (1, 1, 0, -1), \quad \mathbf{v}_2 = (1, 0, 1, -1).$$

Normalize the first vector:

$$\mathbf{u}_1 = \frac{1}{\sqrt{3}}(1, 1, 0, -1).$$

Remove the component of $\mathbf{v}_2$ in the direction of $\mathbf{v}_1$:

$$\mathbf{w}_2 = \mathbf{v}_2 - \frac{\mathbf{v}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 = \mathbf{v}_2 - \frac{2}{3} \mathbf{v}_1 = \frac{1}{3}(1, -2, 3, -1).$$

Use the scaled vector $(1, -2, 3, -1)$ and normalize:

$$\mathbf{u}_2 = \frac{1}{\sqrt{15}}(1, -2, 3, -1).$$

Thus an orthonormal basis for the solution space is

$$\left\{ \frac{1}{\sqrt{3}}(1, 1, 0, -1), \frac{1}{\sqrt{15}}(1, -2, 3, -1) \right\}.$$

The “solution space” here is the **nullspace** of the coefficient matrix: the set of vectors satisfying $A\mathbf{x} = \mathbf{0}$. We first find a nullspace basis and then orthonormalize it.

Closing note

Orthonormal bases simplify coordinates and projections because coefficient extraction requires only inner products.

Definitions and terminology

Term	Meaning
Orthogonal set	Distinct vectors have inner product zero.
Orthonormal set	Orthogonal set of unit vectors.
Orthogonal basis	Basis whose vectors are mutually orthogonal.
Fourier coefficient	Coordinate obtained by inner product with an orthonormal basis vector.

Term	Meaning
Gram-Schmidt process	Converts a basis into an orthogonal or orthonormal basis.
Normalization	Divide a nonzero vector by its norm.
Projection subtraction	Removes components along earlier basis vectors.

Section 5.4 - Mathematical Models and Least Squares Analysis

Learning objectives

Students should be able to:

- formulate a least-squares problem;
- find orthogonal complements and direct sums;
- project onto a subspace;
- identify the four fundamental subspaces;
- solve normal equations;
- construct linear and quadratic least-squares models.

Least-squares problem

Given $A \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$, find $\hat{\mathbf{x}}$ minimizing $\|A\mathbf{x} - \mathbf{b}\|$.

Orthogonal complement

For a subspace $S \subseteq \mathbb{R}^n$, $S^\perp$ is the set of vectors orthogonal to every vector in S .

Orthogonal complement example

Let

$$S = \text{span}\{(1, 1, 0, 1), (1, 0, 1, 1)\}.$$

A vector $\mathbf{x}$ lies in $S^\perp$ when

$$\begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix} \mathbf{x} = \mathbf{0}.$$

A basis for the nullspace is

$$\{(-1, 1, 1, 0), (-1, 0, 0, 1)\}.$$

Therefore,

$$S^\perp = \text{span}\{(-1, 1, 1, 0), (-1, 0, 0, 1)\}.$$

Projection onto a subspace

If $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ is an orthonormal basis for S , then $\text{proj}_S \mathbf{v} = \sum_i \langle \mathbf{v}, \mathbf{u}_i \rangle \mathbf{u}_i$. For an orthogonal basis, divide each term by $\mathbf{w}_i \cdot \mathbf{w}_i$.

Projection onto a plane

Let

$$\mathbf{w}_1 = (1, 1, 0), \quad \mathbf{w}_2 = (1, -1, 2).$$

These vectors are orthogonal. Project

$$\mathbf{v} = (9, 12, 20)$$

onto $S = \text{span}\{\mathbf{w}_1, \mathbf{w}_2\}$:

$$\text{proj}_S \mathbf{v} = \frac{21}{2} \mathbf{w}_1 + \frac{37}{6} \mathbf{w}_2 = \left(\frac{50}{3}, \frac{13}{3}, \frac{37}{3} \right).$$

The residual is

$$\mathbf{v} - \text{proj}_S \mathbf{v} = \frac{23}{3}(-1, 1, 1),$$

which is orthogonal to both spanning vectors.

The four fundamental subspaces

For

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 3 & 1 & 2 \end{bmatrix},$$

the third row equals the sum of the first two. The RREF is

$$\begin{bmatrix} 1 & 0 & -2 & -1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Therefore:

- $\text{rank}(A) = 2$;
- a row-space basis is $\{(1, 0, -2, -1), (0, 1, 1, 1)\}$;
- a column-space basis is formed from the first two original columns;
- a nullspace basis is $\{(2, -1, 1, 0), (1, -1, 0, 1)\}$;
- a basis for $N(A^T)$ is $\{(-1, -1, 1)\}$.

Formal overview

For an $m \times n$ matrix A :

$$\text{Col}(A) \subseteq \mathbb{R}^m, \quad \text{Null}(A) \subseteq \mathbb{R}^n,$$

$$\text{Row}(A) = \text{Col}(A^T) \subseteq \mathbb{R}^n, \quad \text{Null}(A^T) \subseteq \mathbb{R}^m.$$

Moreover,

$$\text{Row}(A)^\perp = \text{Null}(A), \quad \text{Col}(A)^\perp = \text{Null}(A^T).$$

Normal equations

A least-squares solution satisfies $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$. The fitted vector $A \hat{\mathbf{x}}$ is the projection of $\mathbf{b}$ onto $\text{Col}(A)$.

Driving-data regression line

A fictional driving dataset uses

$$(1, 2), (2, 3), (3, 5), (4, 4), (5, 7).$$

Fit

$$y = c_0 + c_1x.$$

The design matrix and data vector are

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 2 \\ 3 \\ 5 \\ 4 \\ 7 \end{bmatrix}.$$

The normal equations are

$$A^T A \begin{bmatrix} c_0 \\ c_1 \end{bmatrix} = A^T \mathbf{b}.$$

Compute

$$A^T A = \begin{bmatrix} 5 & 15 \\ 15 & 55 \end{bmatrix}, \quad A^T \mathbf{b} = \begin{bmatrix} 21 \\ 75 \end{bmatrix}.$$

Solving gives

$$c_0 = \frac{9}{10}, \quad c_1 = \frac{11}{10}.$$

Thus

$$\hat{y} = 0.9 + 1.1x.$$

Quadratic travel model

Suppose a fictional travel-planning response is observed at

$$(0, 3), (1, 6), (2, 11), (3, 18), (4, 28).$$

Fit

$$y = c_0 + c_1x + c_2x^2.$$

Solving the quadratic normal equations gives

$$\hat{y} = \frac{108}{35} + \frac{57}{35}x + \frac{8}{7}x^2.$$

The residual sum of squares is

Closing note

Least squares replaces an inconsistent equation by an orthogonal projection problem. The residual must be orthogonal to the column space.

Definitions and terminology

Term	Meaning
Least-squares solution	Minimizes $\ A\mathbf{x} - \mathbf{b}\ $.
Orthogonal complement	Vectors orthogonal to every vector in a subspace.
Direct sum	Unique decomposition into vectors from two subspaces.
Projection onto a subspace	Closest vector in that subspace.
Four fundamental subspaces	$\text{Col}(A)$, $N(A)$, $\text{Col}(A^T)$, and $N(A^T)$.
Normal equations	$A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$.
Residual	$\mathbf{b} - A\hat{\mathbf{x}}$.
Regression model	Least-squares model for observed data.

Section 5.5 - Applications of Inner Product Spaces

Learning objectives

Students should be able to:

- compute and interpret the cross product in $\mathbb{R}^3$;
- find vectors orthogonal to two given vectors;
- use cross products to find area and angle;
- construct continuous least-squares approximations;
- compute finite Fourier approximations.

Definition: cross product

For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^3$, the vector $\mathbf{u} \times \mathbf{v}$ is orthogonal to both vectors and has magnitude $\|\mathbf{u}\|\|\mathbf{v}\| \sin \theta$.

Hiking-trail cross product

Let two trail-direction vectors be

$$\mathbf{u} = (4, 1, 2), \quad \mathbf{v} = (1, 3, -1).$$

Then

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & 1 & 2 \\ 1 & 3 & -1 \end{vmatrix} = (-7, 6, 11).$$

Check orthogonality:

$$(-7, 6, 11) \cdot (4, 1, 2) = 0,$$

$$(-7, 6, 11) \cdot (1, 3, -1) = 0.$$

The parallelogram area is

$$\|\mathbf{u} \times \mathbf{v}\| = \sqrt{206}.$$

The triangle area is

$$\frac{\sqrt{206}}{2}.$$

Unit normal vectors

A unit vector orthogonal to both $\mathbf{u}$ and $\mathbf{v}$ is

$$\frac{1}{\sqrt{206}}(-7, 6, 11).$$

Its negative is the other unit normal.

Continuous least squares

For an inner product $\langle f, g \rangle = \int_a^b f(x)g(x) dx$, the least-squares approximation is the orthogonal projection of f onto a finite-dimensional function subspace.

Best linear approximation to $f(x) = x^2$ on $[0, 1]$

Approximate $f(x) = x^2$ by

$$g(x) = a + bx.$$

The error must be orthogonal to both basis functions 1 and x :

$$\langle x^2 - a - bx, 1 \rangle = 0,$$

$$\langle x^2 - a - bx, x \rangle = 0.$$

Therefore,

$$\begin{aligned} \int_0^1 (x^2 - a - bx) dx &= 0, \\ \int_0^1 x(x^2 - a - bx) dx &= 0. \end{aligned}$$

This gives

$$\begin{aligned} a + \frac{1}{2}b &= \frac{1}{3}, \\ \frac{1}{2}a + \frac{1}{3}b &= \frac{1}{4}. \end{aligned}$$

Solving,

$$a = -\frac{1}{6}, \quad b = 1.$$

Hence

$$\boxed{g(x) = x - \frac{1}{6}}.$$

Fourier approximation

On $[0, 2\pi]$, consider the centered ramp

$$f(x) = x - \pi.$$

Its Fourier coefficients satisfy

$$a_0 = 0, \quad a_n = 0, \quad b_n = -\frac{2}{n}.$$

The third-order approximation is

$$S_3(x) = -2 \sin x - \sin(2x) - \frac{2}{3} \sin(3x).$$

Closing note

Cross products solve three-dimensional perpendicularity problems, while function-space projections create best approximations and Fourier models.

Definitions and terminology

Term	Meaning
Cross product	Vector orthogonal to two vectors in $\mathbb{R}^3$.
Unit normal	Unit vector perpendicular to a plane.
Parallelogram area	$\ \mathbf{u} \times \mathbf{v}\ $.
Continuous least squares	Projection in a function inner product space.
Approximating subspace	Finite-dimensional function space used for approximation.
Fourier coefficients	Coordinates relative to trigonometric orthogonal functions.
Fourier approximation	Finite trigonometric least-squares approximation.
Approximation order	Highest sine or cosine frequency retained.

Chapter 5 synthesis

Inner products turn vector spaces into geometric environments. Once length and orthogonality are available, we can build orthonormal coordinates, projections, least-squares models, cross-product geometry, and Fourier approximations.

Core connections

dot product $\longrightarrow$ orthogonality $\longrightarrow$ projection $\longrightarrow$ least squares.

basis $\longrightarrow$ Gram-Schmidt $\longrightarrow$ orthonormal basis $\longrightarrow$ simple coordinates and approximation.

Projection onto an orthogonal basis

For an **orthonormal** basis $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$,

$$\text{proj}_U(\mathbf{v}) = \sum_{j=1}^k (\mathbf{v} \cdot \mathbf{u}_j) \mathbf{u}_j.$$

If the basis is only **orthogonal**, not normalized, divide by the squared norm of each basis vector:

$$\text{proj}_U(\mathbf{v}) = \sum_{j=1}^k \frac{\mathbf{v} \cdot \mathbf{u}_j}{\mathbf{u}_j \cdot \mathbf{u}_j} \mathbf{u}_j.$$